WEIHAO YU

+86-180-0141-4808   yuweihao@link.cuhk.edu.hk

Education

•The Chinese University of Hong Kong *Aug. 2023 - present*

Ph.D. candidate at Department of Electronic Engineering Supervisor: Prof. Yixuan Yuan

•Shanghai Jiao Tong University *Sept. 2020 - Mar. 2023*

M.Eng., Department of Automation Supervisor: Prof. Jie Yang

•Shanghai Jiao Tong University *Sept. 2016 - July 2020*

B.Eng., Department of Automation (Artificial Intelligence track) Supervisor: Prof. Lisheng Wang

Research Interests

- My current research centers on multimodal coding and agentic systems for high-value, real-world tasks. My work spans three directions: (1) web development, i.e., front-end code generation; (2) image to webdev, multimodal front-end generation where images, videos, or live website URLs serve as input; and (3) VL-GDP, where I build multimodal agents for economically valuable real-world tasks such as 3D modeling and design and CAD. Previously, I worked on unified multimodal models. My long-term goal is to advance toward Artificial General Intelligence (AGI).

Projects

•VL-GDP: Multimodal Agents for High-Value Real-World Tasks *Qwen Team*

Role: Conceptualization, Capability Building, Pipeline & Environment Buildout, and Data Production

- Identified high-value multimodal scenarios and architected the data production environment from the ground up, integrating scaffolds, professional software, and MCP tools to align with real-world workflows for 3D modeling and design and CAD.
- Delivered end-to-end agentic trajectory data, from production to quality control, for 3D modeling and CAD tasks, forming a new capability of the upcoming Qwen models.

•Image-to-WebDev: Multimodal Front-End Generation *Qwen Team*

Role: Capability Building, MCP Tooling, Data Pipeline, and Model Validation

- Synthesized multimodal queries containing images, videos, and live-website URLs (covering both reconstruction and reference-based generation) and produced single-HTML and React-based agentic trajectory data at scale, with rigorous quality assessment and filtering.
- Designed 16 MCP tools, including image generation, image editing, video generation, web exploration, and screenshotting, enabling the VLM to iteratively improve webpage quality; validated data effectiveness via model training and ensured conflict-free merging with other datasets.

•WebDev: Front-End Code Generation *Qwen Team*

Role: Capability Building, Query Synthesis, and Data Pipeline

- Built the end-to-end data pipeline from scratch: diverse web-generation query synthesis, data production for single-HTML pages and React-based agentic trajectories, and quality assessment and filtering.
- Helped build the Qwen series' front-end code generation capability from 0 to 1: Qwen 3.6 established the foundation, and Qwen 3.7 Max reached #1 in China and #3 worldwide on the arena code leaderboard.

•3D/4D Healthcare *SJTU / CUHK*

Role: Conceptualization, Experimentation, Implementation, and Writing

- Reconstructed dynamic 3D CT via Gaussian splatting, enabling precise reconstruction and device-free respiratory-cycle estimation, and leveraged explicit 3D information for intraoperative X-ray registration meeting emergency-surgery precision within 5 minutes of training.
- Developed ToothMaker for LLM-guided panoramic dental radiograph generation, and GeoT for dental point-cloud segmentation using noise transition matrices to model label-noise distribution.
- Designed a tree neural network exploiting airway-tree structure to achieve the first subsegmental-level airway classification, combining transformers with graph-based methods under airway-tree priors.

Internships

•Qwen Team, Alibaba Inc. *Nov. 2025 - Present*

Topic: Multimodal foundation models *Hangzhou, China*

•BAAI Vision *Mar. 2023 - Aug. 2023*

Topic: Human-in-the-loop autonomous driving dataset construction *Beijing, China*

•Tencent YouTu Lab *Jan. 2022 - May 2022*

Topic: Image retrieval *Shanghai, China*

Publications

- **SpecV: Specification Verification for Robust Unified Multimodal Evaluation**
  Weihao Yu, Rongyao Fang, Yuxuan Cai, Linjiang Huang, Yuhuan Yang, Xianwei Zhuang, Junyang Lin, Yixuan Yuan, Shuai Bai
  European Conference on Computer Vision (ECCV 2026)
- **X^2-Gaussian: 4D Radiative Gaussian Splatting for Continuous-time Tomographic Reconstruction**
  Weihao Yu, Yuanhao Cai, Ruyi Zha, Zhiwen Fan, Chenxin Li, Yixuan Yuan.
  International Conference on Computer Vision (ICCV 2025)
- **GaussianReg: Rapid 2D/3D Registration for Emergency Surgery via Explicit 3D Modeling with Gaussian Primitives**
  Weihao Yu, Xiaoqing Guo, Xinyu Liu, Yifan Liu, Hao Zheng, Yawen Huang, Yixuan Yuan.
  International Conference on Computer Vision (ICCV 2025)
- **ToothMaker: Realistic Panoramic Dental Radiograph Generation via Disentangled Control**
  Weihao Yu, Xiaoqing Guo, Wuyang Li, Xinyu Liu, Hui Chen, Yixuan Yuan.
  IEEE Transactions on Medical Imaging (TMI)
- **TNN: Tree neural network for airway anatomical labeling**
  Weihao Yu, Hao Zheng, Yun Gu, Fangfang Xie, Jie Yang, Jiayuan Sun, Guang-Zhong Yang.
  IEEE Transactions on Medical Imaging (TMI)
- **GeoT: Geometry-guided Instance-dependent Transition Matrix for Semi-supervised Tooth Point Cloud Segmentation**
  Weihao Yu, Xiaoqing Guo, Chenxin Li, Yifan Liu, Yixuan Yuan.
  Information Processing in Medical Imaging (IPMI 2025)
- **AirwayFormer: Structure-Aware Boundary-Adaptive Transformers for Airway Anatomical Labeling**
  Weihao Yu, Hao Zheng, Yun Gu, Fangfang Xie, Jiayuan Sun, Jie Yang.
  International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)
- **Break: Bronchi reconstruction by geodesic transformation and skeleton embedding**
  Weihao Yu, Hao Zheng, Minghui Zhang, Hanxiao Zhang, Jiayuan Sun, Jie Yang.
  IEEE International Symposium on Biomedical Imaging (ISBI 2022)